

ADVAIT NAIR

CIVIL ENGINEERING @ UT AUSTIN | STRUCTURES & MECHANICAL DESIGN



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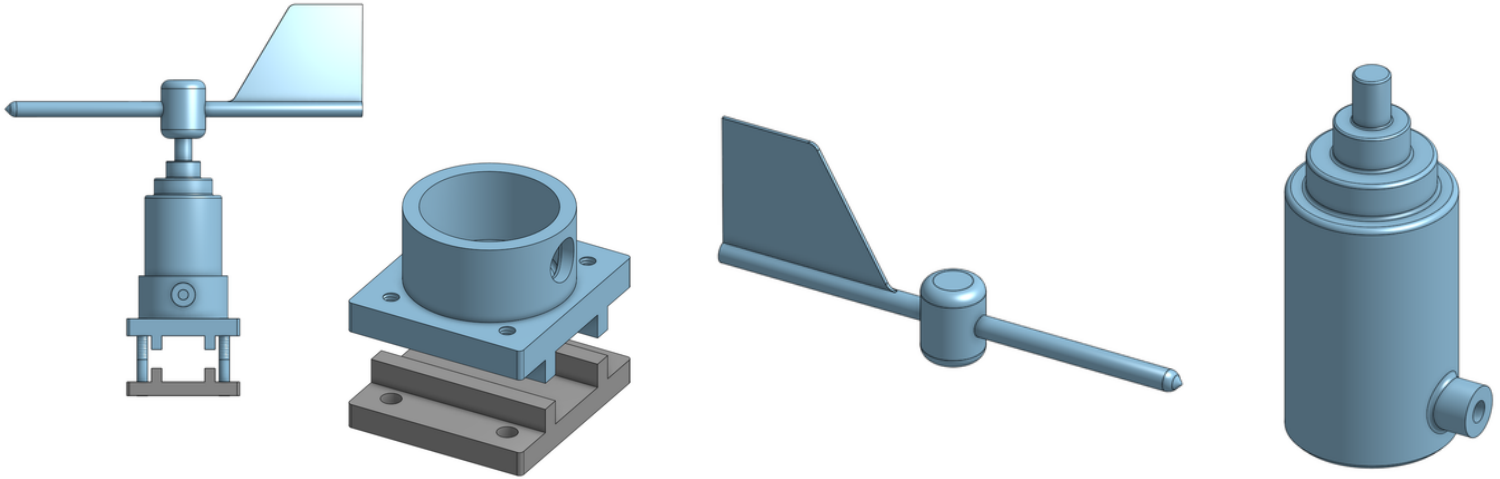


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Austin, Texas

WIND DIRECTION SENSOR MOUNT - SHORELINE AI



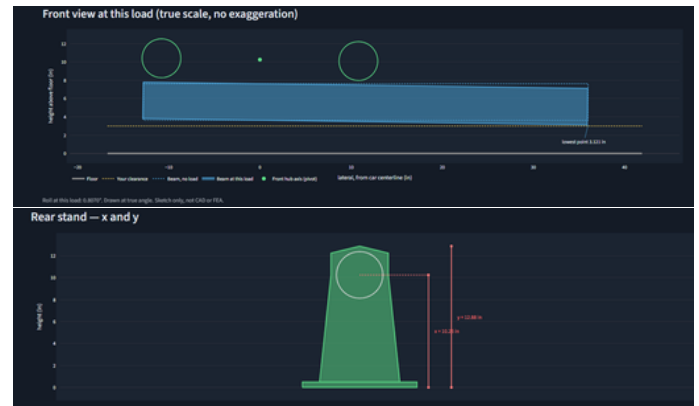
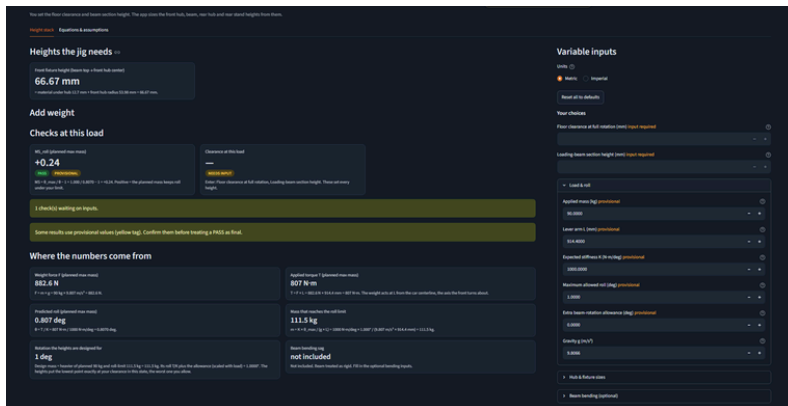
What?

- **Rebuilt** the wind speed and direction sensor as a **CAD model**, working from photographs and measurements
- Design a **mount that clamps the sensor** to an existing steel crossarm **without drilling onto mast**
- Make it printable in-house and serviceable by a field technician with hand tools

How?

- Modelled the sensor in **Onshape** as a stepped housing, rotating vane, and bearing shaft.
- Built the mount as **two bolted plates with lip rails** that fit into the tube's flat faces, **blocking rotation by geometry**
- Selected **ASA filament** based on **Permian Basin field conditions** (heat & wind)

TORSIONAL RIGIDITY JIG CALCULATOR | LONGHORN RACING



What?

- Built a **Python sizing calculator** for the torsional rigidity test jig
- Calculates how tall every part of the jig must be so the load beam can move down under load without hitting the floor
- Includes an add-weight slider that shows the load at which the setup reaches its roll limit and the load at which it contacts the floor

How?

- Followed one causal chain: weight $\rightarrow$ torque $\rightarrow$ roll angle $\rightarrow$ beam rotation $\rightarrow$ hub height, sized for the worst-case load
- Used exact rotation geometry and beam overhang deflection to locate the lowest point at full roll
- Added margin-of-safety checks on roll angle and floor clearance
- Live App Link: <https://tinyurl.com/JigMeasure>

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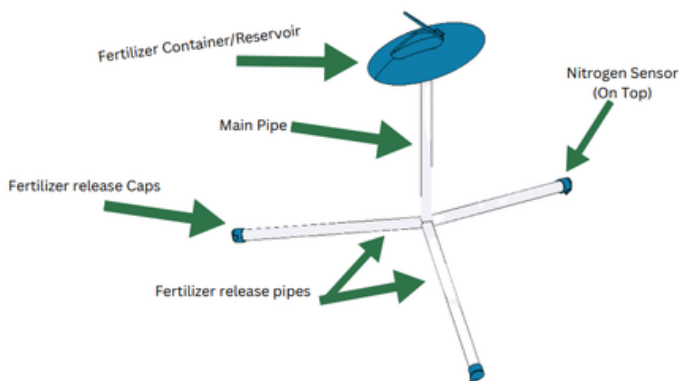


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NITROGENMAPS - ENGINEERING DESIGN @ TECH. STUDENT ASSN.



Photographs of Prototyping Process

Rough Sketch:



Original Prototype:

A system that used standard drip irrigation to distribute liquid fertilizer throughout the field, albeit, in an archaic way, simply dispensing liquid fertilizer onto a patch of soil. This solution didn't seem effective as it poorly distributed nitrogen.



Refinement #1:

With this design, a pump atop a patch of soil would dispense the liquid fertilizer down into the ground, splitting it three ways, more evenly distributing the fertilizer for more efficient and effective use. This also takes away the requirement of a large liquid fertilizer storage tank off site as each unit acts as its own storage tank.

What?

- Designed **NitrogenMAPS**, an automated **soil-nitrogen monitoring** and **precision-fertilization system**
- Delivered fertilizer when soil **nitrogen** dropped below a **target threshold**

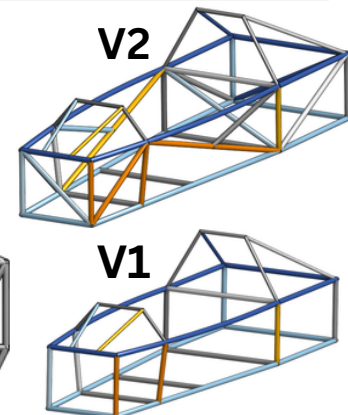
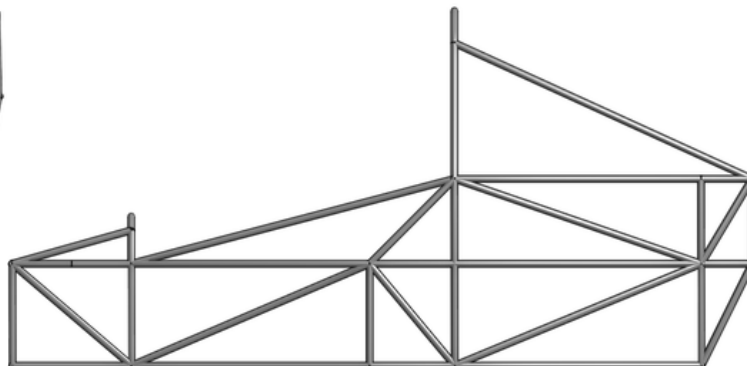
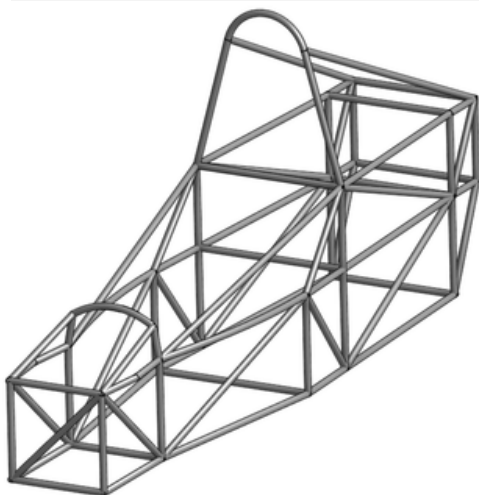
How?

- Integrated an above-ground fertilizer reservoir with a sensor-controlled subsurface distribution network
- Completed five CAD revisions, evolving the design from drip irrigation into a self-contained probe with a pyramidal distribution for wider coverage
- Evaluated cost and adoption barriers and developed a three-level validation plan spanning component, controlled-field, and multi-season trials

Results

- 4th place in Texas TSA State

SPACE FRAME CHASSIS - PERSONAL PROJECT



What?

- Design a **tubular space frame chassis**
- Met **FSAE roll structure requirements** and provided **torsional rigidity**

How?

- Modelled **73 members** in **Onshape** from a **3D sketch skeleton**
- Iterated through **three design revisions**, converting an **unbraced ladder frame** into a **fully triangulated structure**
- Extracted mass and section properties to baseline frame weight and material volume